

Short-Term Price Reversal Following Extreme Negative-Return and High-Volume Events in Major Indian Equities

1. Hypothesis Pitch

This project tests whether extreme negative daily adjusted-close returns accompanied by unusually high relative trading volume are followed by short-term price reversal in major Indian equities. The proposed mechanism is temporary selling pressure or investor overreaction: intense selling may push prices below levels subsequently supported by buying. An alternative explanation is that the decline reflects new adverse information, in which case prices may continue falling.

The hypothesis would receive support from positive subsequent returns that remain economically meaningful after trading costs and portfolio constraints, especially outside the development sample. Weak or negative net returns would count against a useful trading effect. The completed portfolio test produced a 9.51% cumulative return, but its 2.34% annualised return and 0.434 Sharpe ratio provide only modest descriptive support. The methodological limitations below prevent a claim of a robust, implementable anomaly.

2. Data Pipeline

Daily OHLCV data and adjusted closing prices were downloaded from Yahoo Finance using `yfinance` for a fixed universe of 50 major Indian equities, using NSE ticker identifiers. The requested historical window was 1 January 2015 to 1 January 2026, with the end date exclusive. The intended development period was 2015–2021 and the intended evaluation period was 2022–2025. The fixed universe does not reconstruct historical index membership.

The workflow aligned stock series by date, inspected missing observations and listing histories, and checked OHLC consistency. Missing history was not treated as a zero return: adjusted-close percentage changes used `fill_method=None`. Returns and relative volume required observed data, so shorter-history stocks became usable only when sufficient observations existed. The OHLC diagnostic reported no violations, although this did not establish that every opening price was reliable.

Daily return was calculated as adjusted close today divided by adjusted close on the previous trading day, minus one. Relative volume was current volume divided by the mean volume over the previous 20 trading days, with the rolling average shifted by one day. Opening prices were put on a consistent adjusted basis using $\text{adjusted open} = \text{raw open} \times (\text{adjusted close} / \text{raw close})$. These are adjusted-price backtest units rather than literal historical cash execution quotes.

The project saved `raw_market_data.csv` and, after simulation, `oos_portfolio_results.csv`, `oos_executed_trades.csv` and `results_summary.csv` in its data folder. The notebook was reported to run successfully from a restarted kernel. Retaining the downloaded snapshot matters because a later data download may differ from the original research dataset.

3. Strategy and Backtest

A long signal occurs at the close of day t when both the adjusted-close return is at or below the pooled in-sample 5th percentile (approximately -2.9248%) and relative volume is at or above the pooled in-sample 95th percentile (approximately 2.2386 times). These thresholds were learned in-sample and frozen before evaluating the later period. Exploratory in-sample event returns were examined over one, three and five days; the final executable rule used a three-trading-day holding period.

The strategy enters at adjusted open on $t+1$ and exits at adjusted open on $t+4$. Opening prices were flagged when the absolute adjusted open-to-open return exceeded the in-sample 99.9th percentile, approximately 13.43%. Candidate trades were retained only when both entry and exit prices passed this rule and the required observations existed. This retrospective filter has a material limitation discussed in Sections 4 and 5.

The portfolio starts with ₹100,000. New allocations are capped at 10% of portfolio value before that day's entries and constrained by available cash, including entry costs. Total long exposure is funded without leverage; unused funds remain as cash without interest. Fractional shares are allowed. Scheduled exits occur before new entries; a stock already held cannot receive another position. Simultaneous trades follow the existing deterministic order, with allocations reduced to available cash or skipped when no cash remains. Holdings are valued at adjusted opening prices. The 10% limit is an entry-sizing rule, not continuous rebalancing.

The assumed round-trip cost is 0.10%, implemented in the portfolio as 0.05% of purchase value and 0.05% of sale value. The preliminary trade study instead subtracts 0.10 percentage points from gross returns. The final portfolio executes 334 of the 366 retained OOS candidates; candidate returns and executed portfolio returns are therefore different quantities.

Reported results and sample comparison	Result
Initial portfolio value	₹100,000.00
Final portfolio value	₹109,510.22
Cumulative return	9.51%
Annualised return	2.34%
Annualised volatility	5.70%
Sharpe ratio	0.434
Maximum drawdown	-9.86%
Executed trades	334
Win rate	49.70%
Average net trade return	0.29%
In-sample candidate trades / mean net return	590 / -0.1496%
OOS candidate trades / mean net return ¹	366 / $+0.6042\%$

¹ OOS candidate mean net return = 0.7042% gross less the 0.10% flat trade-study cost. These are unconstrained trade averages; no in-sample portfolio return or Sharpe was produced for a like-for-like portfolio comparison.

4. Methodology Defense

The major Indian equity universe provides a manageable setting for investigating a specific price-and-volume hypothesis. Relative volume compares activity with each stock's own recent history, while return percentiles give an explicit definition of an extreme decline. Pooled thresholds make the implementation transparent, although stocks with different volatility distributions need not have equal signal probabilities. Daily observations support a clear timing convention without claiming intraday execution precision.

The chronological split allocates roughly seven years to developing the rule and four later years to evaluating it, preserving temporal order rather than mixing future observations into training. A three-day hold targets a short-lived rebound and keeps the rule simple; it was not established as an optimal horizon. The 0.10% round-trip cost is a transparent non-zero baseline, not a market-calibrated estimate. If it understates stressed execution costs, the small apparent edge could disappear. The choice of Indian equities also reflects my interest in understanding a market relevant to my future work, rather than selecting a country after comparing backtest profits.

Separating the close-based signal from next-day opening execution prevents the most direct timing error: buying at an opening price using information that only becomes available at that day's close. Lagging the volume average also prevents the event's own volume from defining its baseline. Adjusting opening and closing prices consistently reduces corporate-action inconsistencies, but does not independently validate the data vendor's observations.

Opening-price diagnostics were important because an extreme ADANIENT observation remained unusual after adjustment and passed basic OHLC checks. The chosen filter applies the same in-sample-derived threshold across stocks rather than deleting one named trade. However, a large move is not proof of a data error. Genuine distressed-market observations may also be excluded, and an independent source check was not established by the completed work.

The candidate list uses the future exit-price flag to retain or exclude an entire trade. That flag is unknown at entry, so freezing the threshold does not remove retrospective selection. The result is conditional on retained observations, rather than a fully causal simulation. A future revision must distinguish verified data repairs from genuine extreme prices and define executable handling of positions when problematic prices arise.

There is also a split-boundary qualification. Although the intended OOS period starts on 1 January 2022, the final signal and portfolio slices shown in the project use `.loc[split_date:]` with `split_date` set to 31 December 2021. This inclusive slice overlaps the in-sample boundary. In-sample forward-return calculations can also extend beyond the split for signals near its end. The reported numbers are preserved, but a strictly isolated test would require a disjoint date boundary and outcomes confined to the relevant evaluation window. The effect of these corrections has not been measured.

Metrics use 989 portfolio dates and 252 trading days per year. Annualised return compounds the initial-to-final value ratio over 989/252 years. Annualised volatility is daily return standard deviation $\times \sqrt{252}$. Sharpe is daily mean return divided by its standard deviation $\times \sqrt{252}$, with zero risk-free return. Drawdown is portfolio value divided by its running peak, minus one. The initial daily return is set to zero.

A win rate of 49.70% can coexist with a positive average return because the sizes of gains and losses matter. Portfolio performance further depends on position sizes, overlapping opportunities and cash. No benchmark-adjusted alpha, formal significance test, execution-filter sensitivity result or walk-forward validation is established by the completed outputs. Positive cumulative performance alone does not establish that the signal outperforms passive exposure or that abnormal volume adds predictive value.

5. Pitfall Checklist

Look-ahead bias

Signals use adjusted-close returns and current volume known by the close, with a lagged 20-day volume baseline and next-day opening entry. These choices protect signal timing. However, excluding a trade because of its future exit-price flag creates retrospective selection. The inclusive sample boundary and forward outcomes crossing that boundary further weaken strict separation. These limitations remain in the reported results; the backtest cannot be described as entirely free of look-ahead bias.

Survivorship bias

The fixed set of major Indian equities was applied to historical data without rebuilding the universe through time. Companies removed from an index, delisted or otherwise absent from the selected list may be omitted, while later constituents may be represented before membership. Allowing shorter-history stocks to enter only when data become available does not resolve membership selection. The results describe this selected universe, not a survivorship-bias-free historical NIFTY strategy. Historical constituent and delisting data would be needed to address this properly. Including omitted distressed or delisted firms could worsen reversal returns by adding persistent declines and difficult exits, but the direction and magnitude cannot be established without those missing histories.

Overfitting/data-snooping

Return and volume thresholds, holding period, costs and the execution filter were recorded as fixed before the later evaluation. The weak in-sample mean was retained rather than forcing an attractive result by changing the strategy. Nevertheless, examining several horizons and devising a filter after investigating extreme observations involve research discretion. A single subsequent period does not eliminate selection risk. Any further parameter exploration should be assessed on new held-out data, not presented as validation on the same already-inspected period. The documented exploration used three event-study horizons (one, three and five days) before fixing the three-day trading rule; no exhaustive parameter grid was recorded. Price-adjustment and filtering revisions were diagnostic changes, so they should not be misrepresented as zero research experimentation.

Transaction costs and slippage

The portfolio deducts 0.05% on each side of a trade, so its reported profit is net of the specified cost model. This is an assumption rather than an empirically calibrated schedule of all market frictions. Explicit opening-auction slippage, bid-ask spreads, market impact and an itemised charge model were not implemented. Stress events can make execution more expensive precisely when the strategy trades. No completed cost-sensitivity analysis establishes how robust the modest return is to higher costs.

Out-of-sample validation

The intended chronological split uses 2015–2021 for development and 2022–2025 for evaluation, and the signal thresholds remain frozen. The final later-period simulation reports 9.51% cumulative return and a Sharpe ratio of 0.434. This is more informative than development-period performance alone, but the boundary overlap, future-exit filtering and fixed universe qualify its independence and realism. The results are descriptive evidence from one historical evaluation, not proof of generalisation across regimes.

6. Efficiency Touchpoint

The sequential portfolio simulation is the expected computational bottleneck in local processing, although no runtime profiling was performed. For T dates, N stocks and M candidate trades, its worst-case time complexity is $O(TN + M)$: each date requires traversing at most N open positions, and each candidate is considered once. At 989 evaluation dates, 50 stocks and 366 candidates, this is manageable. Network download time is a separate source of elapsed time.

Grouping candidates by entry date avoids scanning all trades every day. `pandas` and `NumPy` vectorise returns, rolling volume and signal masks in approximately $O(TN)$ work for a fixed window. The portfolio loop remains sequential because cash and holdings depend on prior decisions; simply combining independent trade returns would lose those constraints.

7. Risk Touchpoint

New stock allocations are capped at 10% of portfolio value and spending cannot exceed cash after costs. There is no borrowing or short selling. These controls limit initial concentration, but weights can drift above 10% because there is no continuous rebalancing. The completed strategy has no stop-loss or sector cap.

A plausible adverse scenario is a market-wide sell-off in which many stocks simultaneously meet the negative-return and high-volume conditions. The model may buy several correlated stocks while the decline reflects lasting deterioration. Positions can continue falling through the fixed holding period, with overnight gaps and worse opening execution adding losses. A short holding period does not eliminate tail risk, and ten nominally different stocks need not provide ten independent sources of risk.

A -9.86% maximum drawdown is substantial relative to the 2.34% annualised return. The 5.70% volatility includes cash and does not imply low individual-trade risk. Opening-price valuation omits intraday worst-case losses, while a zero-risk-free Sharpe and no cash interest simplify the opportunity-cost comparison.

8. Reflection

I learned that a plausible reversal story and a high proportion of winning trades are not enough to establish a useful strategy. The working hypothesis anticipated temporary selling pressure and a rebound, yet the cleaned in-sample average was negative despite more than half of gross trades being profitable. This challenged the assumption that frequent rebounds would outweigh continued declines. Building a funded portfolio also showed how execution timing, overlapping holdings, costs and cash change the result. Debugging made clear that code running without errors does not establish that its financial interpretation is correct.

The final reported portfolio gained ₹9,510.22, but the low annualised return, 0.434 Sharpe ratio and near-10% drawdown do not establish a strong trading anomaly. The negative in-sample average and positive later performance suggest that results vary across periods; they do not identify a stable causal reversal mechanism. Survivorship bias and retrospective execution filtering further limit confidence.

My next steps would be to reconstruct historical membership, independently validate anomalous prices, handle execution problems causally and make sample boundaries disjoint. Only then would I test costs, passive benchmarks and additional held-out periods. Larger capital or live trading would require realistic lot sizes, liquidity and market-impact models, and rules for unavailable prices or failed orders. These are proposed improvements, not completed tests. The defensible outcome is a modest historical result with explicit limitations.